

LUKE BOLLENBACHER

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Active DOE Q Clearance (Top Secret Level)

EDUCATION

Texas A&M University — B.S. Electrical Engineering Aug 2022 – Dec 2026 (Expected)

College Station, TX • GPA: 3.12/4.0 • Specialties: Device Science & Nanotechnology; Computer Engineering & Systems

- Focus: Quantum and photonic device engineering, electromagnetic systems, computational modeling, and secure electronics for aerospace and defense applications
- Coursework — Devices & EM: Thin Film (440), Lasers (470), EM Fields (322) | Signals & Computing: Random Signals (303), Signals & Systems (314), Machine Learning (427), Computational Data Science (360), Computer Architecture (350), Microprocessor Design (449) | Systems: Power Systems (460), Unconditionally Secure Electronics (466)

EXPERIENCE

PCB Designer Mar 2026 – May 2026

Texas A&M University College of Engineering — College Station, TX — Part-time

- Designed PCBs under Prof. Dr. Lin using Altium Designer for embedded chip systems targeting photonic, quantum, and biological domain applications
- Applied ASIC design principles and signal integrity practices to research-grade boards in a cutting-edge multidisciplinary lab environment

Electrical Engineering CAD/PCB Design Intern Dec 2024 – Feb 2026

Sandia National Laboratories — Livermore, CA — Part- & Full-time

- Performed CAD and PCB design in support of classified electrical engineering programs: schematic capture, component selection, footprint verification, and full PCB layout from system requirements through fabrication-ready deliverables
- Supported design reviews and iterative revisions with senior engineers, incorporating signal integrity, manufacturability, and documentation control under strict configuration management and security protocols
- Obtained TSI//RD Q-level (Top Secret) onsite badge clearance; contributed directly to NNSA-funded national security programs

National Science Foundation R&D Intern Jun 2022 – Jul 2023

Sabin Lab & CornellTech, Cornell University — Ithaca, NY

- Conducted trans-disciplinary design research integrating computational modeling, materials science, mechanical engineering, and architectural design to develop responsive material systems; contributed to NSF-, Autodesk-, Grainger-, and NAE-funded projects including Kirigami 3.0, RoboSense 2.0, and PolyBrick 2.0
- Executed digital fabrication and large-format 3D printing workflows using Rhinoceros3D (v7), Simplify3D, and Potterbot; implemented ML algorithms in C, C++, C#, and Python for circuitry, solar array design, and power distribution applications

TECHNICAL SKILLS

Circuits & Systems: Analog, digital & quantum circuits; biasing, small-signal models; filters; FSMs; quantum devices

Signals & Systems: Fourier/Laplace analysis; LTI systems; spectral density; noise modeling; uncertainty quantification

EM & Device Physics: Maxwell's equations; EM propagation; radiation-matter interaction; semiconductor transport; photonic & quantum devices; thin-film technology

Materials Science: Defects & dopants; Fick diffusion; alloy composition; slip/deformation; grain strengthening & growth

Hardware & HDL: PCB schematic review; FPGA; ALUs; ASIC concepts; Verilog, VHDL; Altium Designer; Siemens Expedition

Simulation & Modeling: COMSOL, LTspice, Quantum ESPRESSO, Geant4, SolidWorks

Programming: Python, Julia, MATLAB, C/C++, ArmV8, Qiskit

Lab & Dev Tools: LabVIEW, VS Code

ACTIVITIES & ORGANIZATIONS

IEEE — Institute of Electrical & Electronics Engineers Aug 2023 – Present

- In-field speakers, professional development workshops, and annual career fair participation

QuantaA&M — Texas A&M Quantum Computing Club Aug 2024 – Present

- Quantum computing industry speakers, book club, and Qiskit quantum coding workshops

TAMU SEDS — Students for Exploration and Development of Space Aug 2024 – Jan 2025

- NASA Lunabotics electrical subteam; attended SpaceVision 2025 and engaged aerospace and defense industry executives

Aggie Coding Club Aug 2022 – Jan 2025

- Personal and team-oriented computer science and embedded systems projects